DIVIDE AND CONQUER METHOD

- ✓ A divide-and-conquer algorithm works by recursively breaking down a problem into two or more sub-problems of the same or related type, until these become simple enough to be solved directly.
- ✓ The solutions to the sub-problems are then combined to give a solution to the original problem.
- ✓ This technique can be divided into the following three parts:

I. **Divide:** This involves dividing the problem into some sub problem.

II. **Conquer:** Solve the Sub problem by calling recursively until all the sub problems solved.

III. **Combine:** The Sub problem solved so that we will get find problem solution.

Divide/Break

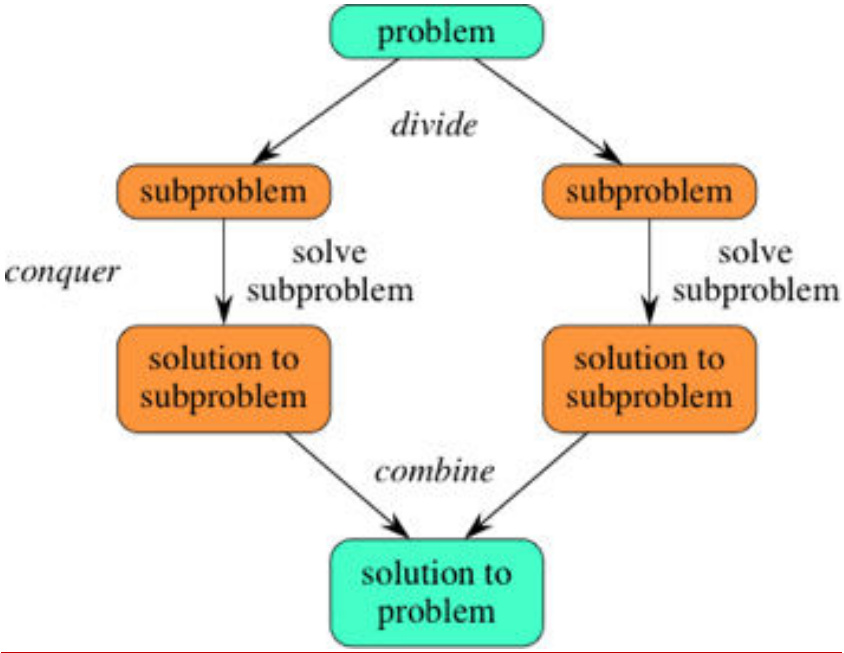
- This step involves breaking the problem into smaller sub-problems.
- Sub-problems should represent a part of the original problem.
- This step generally takes a recursive approach to divide the problem until no sub-problem is further divisible.
- At this stage, sub-problems become atomic in nature but still represent some part of the actual problem.

Conquer/Solve

- This step receives a lot of smaller sub-problems to be solved.
- Generally, at this level, the problems are considered 'solved' on their own.

Merge/Combine

- When the smaller sub-problems are solved, this stage recursively combines them until they formulate a solution of the original problem.



CONTROL ABSTRACTION OF DIVIDE AND CONQUER

- A control abstraction is a procedure whose flow of control is clear but whose primary operations are specified by other procedures whose precise meanings are left undefined.
- The control abstraction for divide and conquer technique is DANDC(P), where P is the problem to be solved.

DANDC (P)

{

if SMALL (P) then return S (P);

else

{

Divide P into smaller instances P_1, P_2, …. P_k, k≥1

Apply DANDC to each of these sub problems;

Return (COMBINE (DANDC (P_1), DANDC (P_2),…., DANDC (P_k)) ;

}

}

- ✓ SMALL(P) is a Boolean valued function which determines whether the input size is small enough so that the answer can be computed without splitting.
- ✓ If this is so function 'S' is invoked otherwise, the problem 'P' is divided into smaller sub problems.
- ✓ These sub problems P_1, P_2, . . . , P_k are solved by recursive application of DANDC.